26900195 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEECT307 - CYBER SECURITY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Identify the act of gaining unauthorized access to a system. [K] / [CO1]

   a). Backup b). Hacking

   c). Encryption d). Scanning

2. Which of the following best distinguishes cyber terrorism from ordinary cybercrime? [U] / [CO2]

   a). Uses Internet and computers

   b). Political, ideological, or religious motivation combined with disruption

   c). Unauthorized access

   d). Digital arrests for money extortion

3. What is DNS spoofing? [U] / [CO3]

   a). Increasing mapping speed

   b). Encrypting DNS records

c). Providing false DNS information to redirect traffic

d). Increasing DNS server capacity

4. What is the purpose of a PIN on a mobile phone? [A] / [CO4]

a). To authenticate the user b). To increase network bandwidth

c). To improve camera quality d). To update the operating system

5. Name the component that ensures systems and data are accessible when needed. [U] / [CO1]

a). Integrity b). Availability

c). Confidentiality d). Encryption

6. Which legal principle is particularly relevant when digital evidence is used in court? [A] / [CO2]

a). Authenticity and integrity of evidence

b). High resolution videos and images

c). Recent images

d). Digital Signatures

7. What is a major weakness of transmitting sensitive information over an unencrypted connection? [U] / [CO1, CO3]

a). High packet payload

b). Shorter time to live

c). Transmission is slowed down

d). It may be intercepted and read

8. Which of the following presents a significant risk to a web server? [A] / [CO3]

a). Privileged access b). Unpatched software

c). No regular backups d). Strong authentication

9. Name the layer that protects information from unauthorized access. [U] / [CO1]

a). Physical security b). Network security

c). Data security d). Application security

10. Which factor can make cybercriminal behaviour more difficult to detect? [U] / [CO2]

a). Public disclosure of all activities

b). Physical presence at the victim's location

c). Use of one's legal identity in every activity

d). Use of aliases and anonymization techniques

11. Which attack can execute malicious scripts in a victim's browser? [U] / [CO3]

a). DDoS attack b). Cross-Site Scripting

c). DNS amplification d). Web jacking

12. A fraudulent SMS containing a malicious link is commonly associated with: [A] / [CO4]

a). Authentication b). Encryption

c). Hashing d). Smishing

13. Name the model that focuses on data integrity. [U] / [CO1]

a). Brewer-Nash b). Clark-Wilson Model

c). Biba d). Take-Grant

14. Which organization is designated as India's national nodal agency for responding to cybersecurity incidents? [K] / [CO2]

a). NCIIPC b). UIDAI c). CERT-In d). TRAI

15. Which of the following is a common characteristic of spyware? [K] / [CO1, CO3]

a). It prevents all unauthorized access

b). Stops all services

c). Modifies databases

d). It secretly monitors user activities

16. Which factor has significantly contributed to the growth of mobile computing? [K] / [CO4]

a). Decline in wireless networks b). Increasing use of smartphones

c). Reduction in Internet usage d). Elimination of mobile applications

[1x16 = 16]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

17. Explain the common software attacks and state suitable preventive measures. [U] / [CO1]

18. Explain the difference between intentional and unintentional cyber threats with examples [K] / [CO2]

19. What are the features introduced in http/2 to assure improved performance? [K] / [CO3]

20. Why is user education considered a critical component of mobile security policy? [A] / [CO4]

21. Describe the different types of attacks and give suitable examples. [U] / [CO1]

22. Explain cyber terrorism and distinguish it from ordinary cyber crime [A] / [CO2]

23. What is hashing? How is it different from encryption? [U] / [CO3]

24. Why should organizations maintain a corporate asset register for all mobile devices, including employee-owned ones? [An] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **12** marks

25. Discuss the major security challenges associated with mobile devices. Explain the risks arising from device loss or theft, insecure networks, malicious applications, outdated software, and excessive application permissions. [U] / [CO4]

26. Discuss password-cracking techniques, commonly used tools, and methods for preventing password-cracking attacks. [U] / [CO1]

27. Explain the scope and objectives of cyber law. Also discuss the relationship between cybersecurity regulations and organizational security practices. [U] / [CO2]

28. Explain the major components of the Internet and describe how they interact to provide Internet services [K] / [CO3]

[12x2 = 24]